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Recent developments in Zn-catalyzed asymmetric addition reaction to ketones: Syntheses of chiral tertiary alcohols

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Abstract:

Zn-catalyzed asymmetric alkylation reaction with carbonyl compounds has provided a straightforward method to access enantio-enriched tertiary alcohols. This chiral carbon-carbon bond forming process has attracted great attention in the field of asymmetric synthesis with features of mild reaction conditions, tolerating diverse functional groups and employment of low toxic zinc metal. In this digest, the recent developments of this asymmetric protocol are focused.

Introduction

Catalytic asymmetric addition of organozincs to aldehydes has been considered as a straightforward route to access optically active alcohols since its arrival. This method, pioneered by Oguni and Omi¹ and subsequently developed by Noyori and coworkers for introducing efficient DAIB ligand [3-*exo*-(dimethylamino)isoborneol],² have stimulated the interests of designing and synthesizing novel chiral modifiers to study their catalytic reactivity and enantioselectivity. The origin of high enantioselectivity for the asymmetric addition reaction of a diorganozinc reagent to an aromatic aldehyde using a β -amino alcohol ligand proposed by Noyori and coworkers is illustrated in Figure 1.³ Initially, complex A for the enantioselective reaction is formed preferentially by producing alkane and subsequent coordination with another molecule of dialkylzinc via O-atom transforms complex A into complex B, which the Zn-atom is now acting as Lewis acid to activate aldehyde to form complex C. In the four-membered transition structure D, alkyl migration to the *Si* face of carbonyl group was calculated as an energetically favorable pathway, giving rise to complex E, which

the resulting zinc alkoxide exchanges with dialkylzinc to regenerate complex B (Figure 1). This asymmetric carbon-carbon forming process has been extended to catalytic reactions of ketones and ketoesters, challenging substrates for the preparation of chiral tertiary alcohols due to its tolerance of wide range of substituents, mild reaction conditions and the utilization of a rather less-toxic Zn metal. While numerous reviews collectively address asymmetric Zn-catalyzed addition of aldehydes,⁴ this digest focuses the recent developments, within the last ten year, on the syntheses of chiral alcohols with quaternary stereogenic centers from Zn-catalyzed asymmetric addition reaction of ketones and ketoesters.

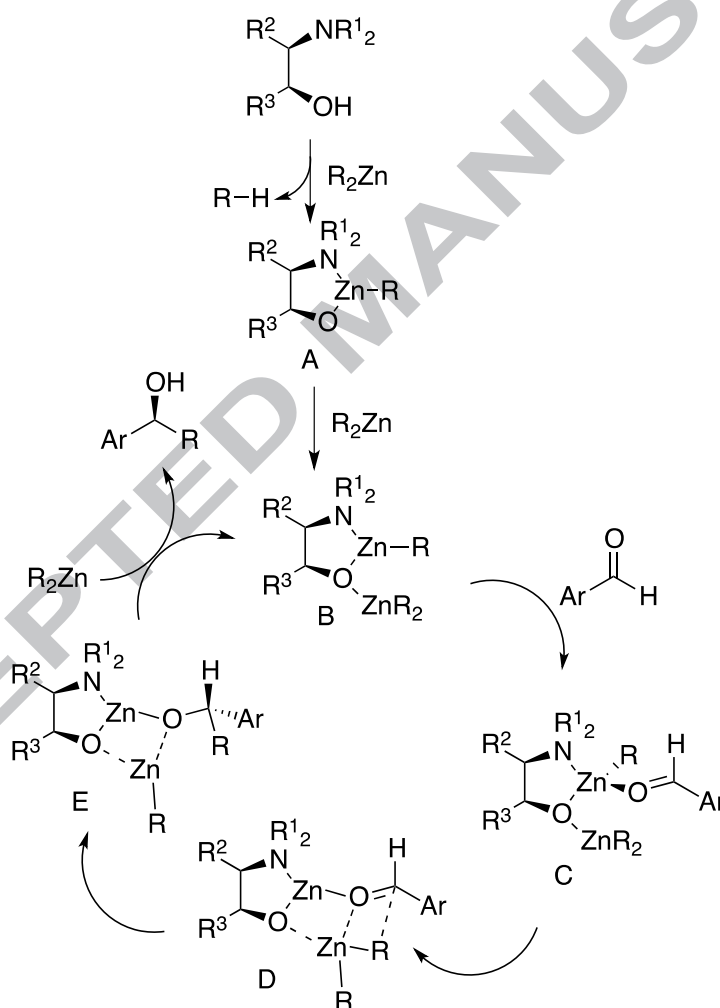
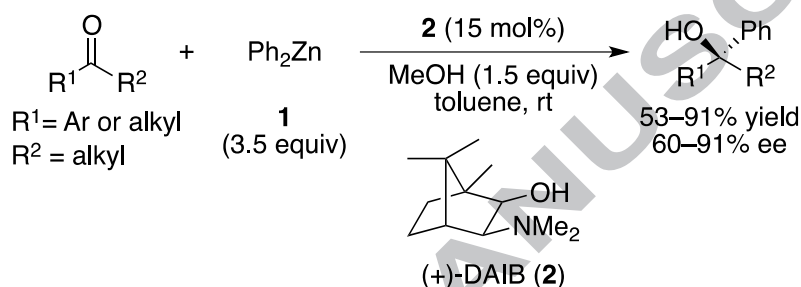


Figure 1. Proposed mechanism for asymmetric dialkylzinc addition to ArCHO

The addition of arylzincs and dialkylzincs to aromatic ketones

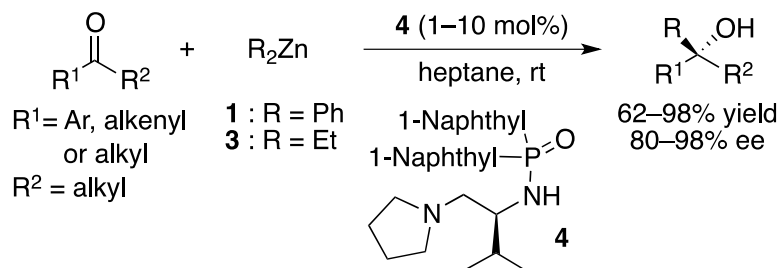
Ketones, as opposed to aldehydes, are less reactive electrophiles toward nucleophilic addition of organozinc reagents, giving rise to a recovery of the unreacted materials or

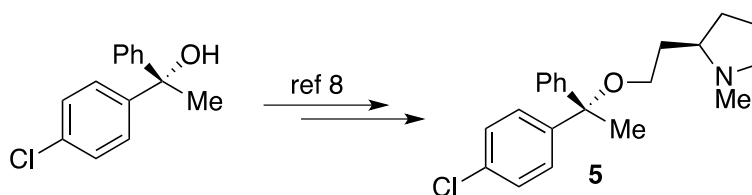
the formation of side products due to reduction.⁵ Breakthrough was made when Fu and coworker conducted the catalytic enantioselective addition of aryl-alkyl and dialkyl ketones with diphenylzinc (**1**) employing 15 mol% of (+)-DAIB (**2**) (Scheme 1).⁶ The corresponding tertiary alcohols were obtained in up to 91% yield and 91% ee. In this method, improved yields and enhanced enantioselectivities were witnessed when excess **1** (3.5 equiv) along with MeOH (1.5 equiv) were added. Replacing a phenyl group with a methoxy group on the zinc complex enhances the Lewis acidity of the chiral catalyst and facilitates the reaction of diphenylzinc and ketones.



Scheme 1. Asymmetric phenylation of ketones in the presence of (+)-DAIB (**2**).

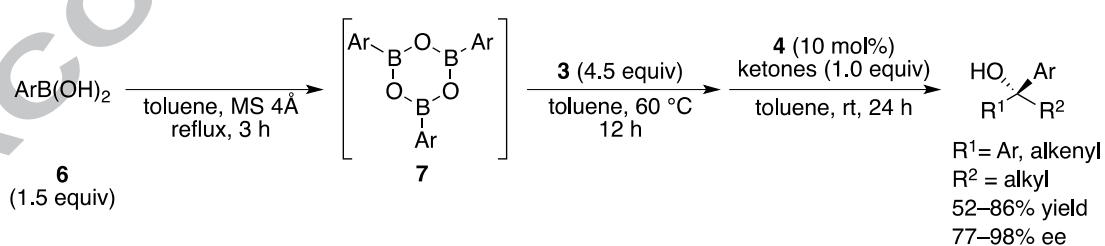
Ishihara and coworkers subsequently reported that catalytic addition of Ph_2Zn (**1**) and Et_2Zn (**3**) to ketones, in the presence of chiral phosphoramidate ligands that derived from natural L-valine, proceeded in an efficient and highly stereoselective manner under mild^{7a} or solvent free conditions^{7b} at room temperature to offer the corresponding tertiary alcohols with up to 98% yield and 99% ee (Scheme 2).⁷ This catalytic system is applicable to a variety of aromatic and aliphatic ketones with the Zn(II)-phosphoramidate **4** acting as the conjugate Lewis acid-Lewis base catalyst during the catalytic cycle. A formal synthesis of clemastine (**5**) was demonstrated by means of this method.



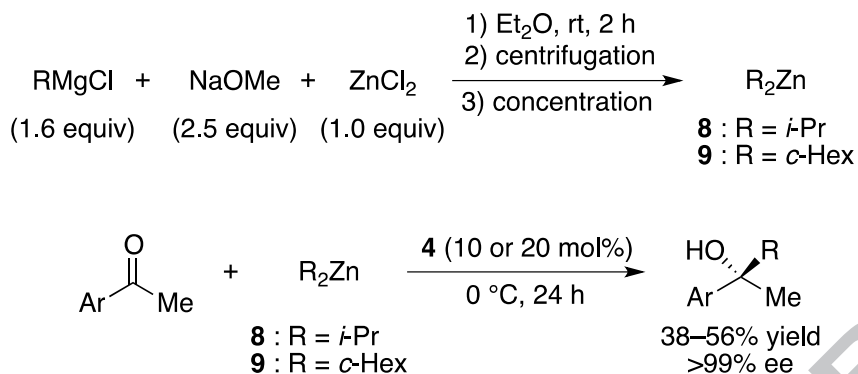


Scheme 2. Catalytic addition of organozincs with ketones in the presence of **4** and formal synthesis of clemastine **5**.

More practically, catalytic asymmetric aryl or alkyl addition to ketones with organozinc reagents that generated *in situ* from arylboronic acids and alkyl Grignard reagents, respectively, was later reported by Ishihara and coworkers.⁹ While arylzincs could be prepared employing Bolm's protocol,¹⁰ which generally requires 2.4 equiv of arylboronic acids **6** and 7.2 equiv of diethylzinc (**3**). Replacement of arylboronic acids with their triarylboroxine derivatives **7** reduces the required amount of **3** to 4.5 equiv (Scheme 3).¹¹ The enantioselective arylation reaction of ketones employing thus obtained arylzinc reagents in the presence of 10 mol% of phosphoramidate **4** provides the corresponding addition products in up to 86% yield with 98% ee. In the same report, salt-free (*i*-Pr)₂Zn (**8**) and (*c*-Hex)₂Zn (**9**) reagents,¹² prepared from the corresponding Grignard reagents, ZnCl₂ and NaOMe, were utilized in the catalytic enantioselective alkylation reaction of ketones, furnishing the corresponding tertiary alcohols in 56% yield with >99% ee (Scheme 4).

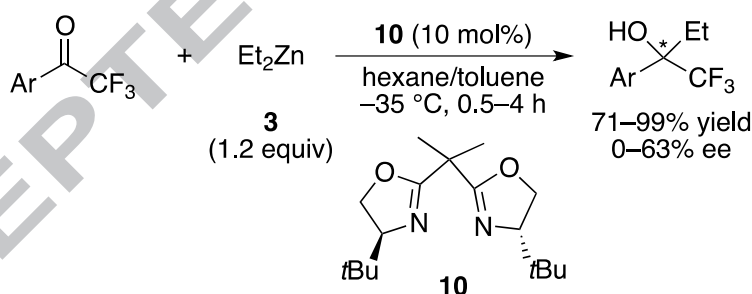


Scheme 3. Catalytic enantioselective addition of arylzincs with ketones in the presence of **4**.

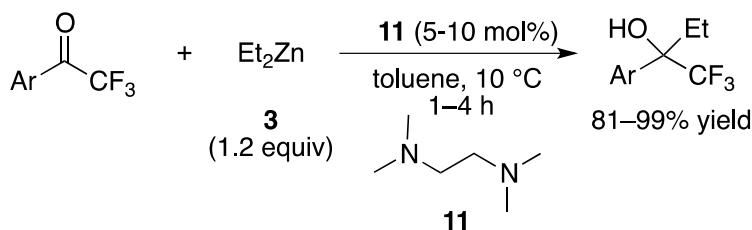


Scheme 4. Catalytic enantioselective addition of (*i*-Pr)₂Zn (**8**) and (*c*-Hex)₂Zn (**9**) with ketones in the presence of **4**.

Wolf and coworker disclosed that asymmetric Et₂Zn (**3**) addition to 2,2,2-trifluoromethyl arylketones utilizing 10 mol% of TBOX (**10**) as ligand, giving rise to the corresponding tertiary alcohols in up to 99% yield and 63% ee (Scheme 5).¹³ They also found that racemic tertiary alcohols could be prepared with up to 99% yield when the addition reactions were catalyzed in the presence of 10 mol% of TMEDA (**11**) (Scheme 6)

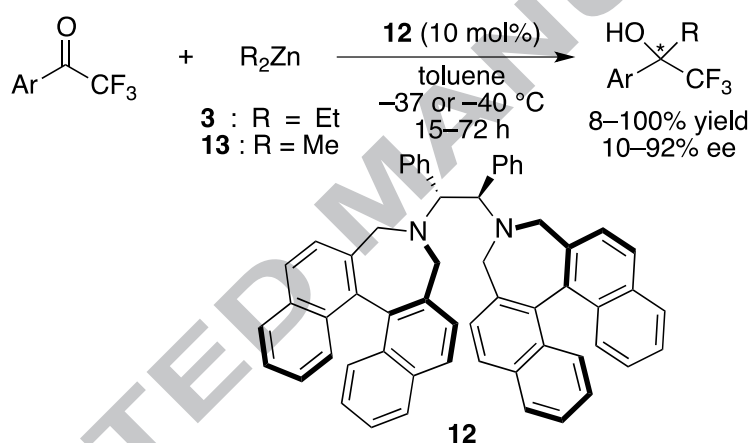


Scheme 5. Catalytic enantioselective addition of Et₂Zn (**3**) to 2,2,2-trifluoromethyl arylketones in the presence of **10**.



Scheme 6. Addition of Et₂Zn (**3**) to 2,2,2-trifluoromethyl arylketones in the presence of **11**.

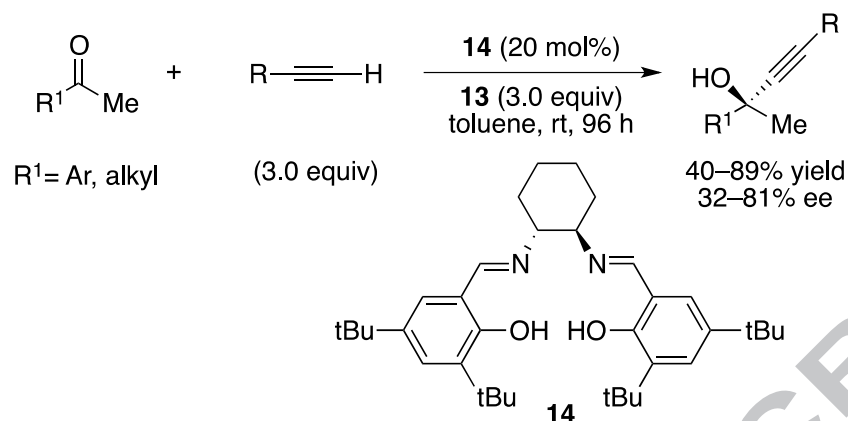
While achiral TMEDA (**11**) ligand was demonstrated to be effective in the production of racemic fluorinated tertiary alcohols, the employment of chiral diamine in the same asymmetric transformation was unknown. It was until Espinet and coworkers described that chiral diamine **12**, derived from (*R,R*)-1,2-diphenylethylenediamine, is an efficient ligand for Et₂Zn (**3**) and Me₂Zn (**13**) addition to 2,2,2-trifluoromethyl arylketones, providing the corresponding tertiary alcohols in up to >99.5% yield with 92% ee (Scheme 7).¹⁴ In a subsequent study, the authors disclosed that chiral diamine ligand **12** was able to autocatalyze with asymmetric amplification in the catalytic Et₂Zn (**3**) addition to 2,2,2-trifluoromethyl arylketones and the reaction rate is inversely dependent on the concentration of catalysts.¹⁵



Scheme 7. Catalytic enantioselective addition of Et₂Zn (**3**) and Me₂Zn (**13**) to 2,2,2-trifluoromethyl arylketones in the presence of **12**.

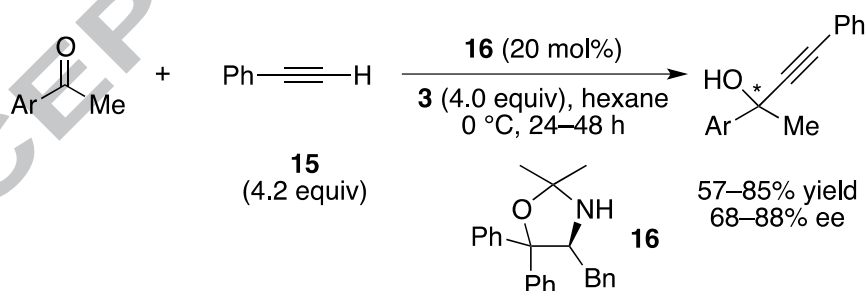
The addition of Zn-acetylides to aromatic ketones

Cozzi has described the first enantioselective alkynylation reaction of ketones with Zn-acetylides, preformed by treating terminal alkynes with Me₂Zn (**13**), in the presence of 20 mol% of commercially available chiral salen ligand **14** comprising (1*R*,2*R*)-1,2-diaminocyclohexane moiety.¹⁶ In this protocol, a large number of aryl and aliphatic ketones were tolerated to give tertiary propargylic alcohols in up to 89% yield and 81% ee (**Scheme 8**).



Scheme 8. Catalytic enantioselective addition of Zn-acetylides to ketones in the presence of **14**.

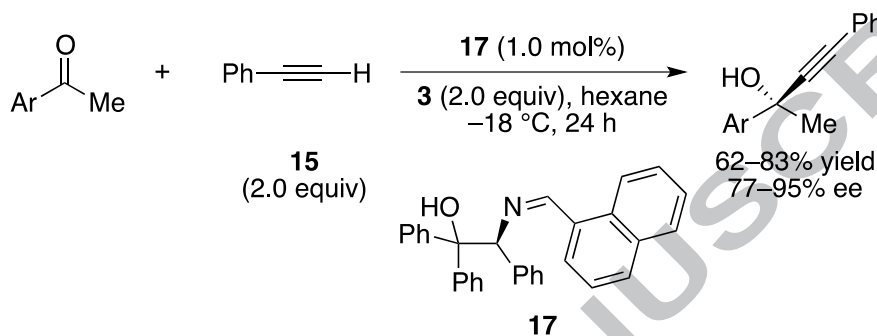
Wang and coworkers employed chiral oxazolidine ligands derived from natural occurring amino acids in the catalytic addition reaction of Zn-phenylacetylide, *in situ* generated from phenylacetylene (**15**) and Et_2Zn (**3**), with ketones. In the presence of 20 mol% of phenylalanine-derived oxazoline ligand **16**, asymmetric addition of Zn-phenylacetylide to a variety of aromatic ketones afforded the corresponding tertiary propargylic alcohols in up to 85% yield and 88% ee (Scheme 9).¹⁷



Scheme 9. Catalytic enantioselective addition of Zn-phenylacetylide to aromatic ketones in the presence of **16**.

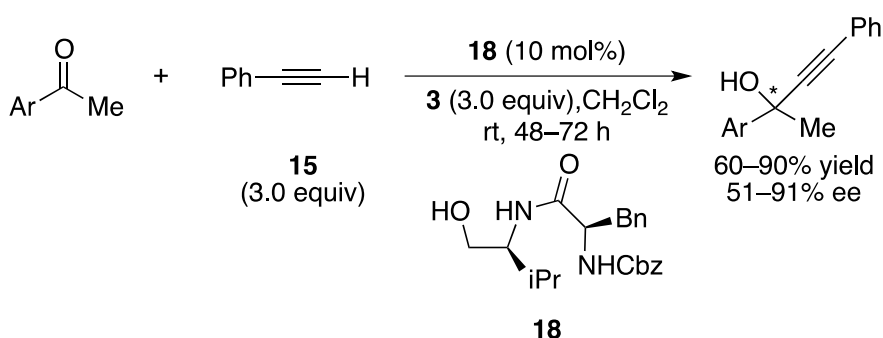
After identifying amino alcohol moieties, crucial fragments for chiral ligands in the syntheses of chiral tertiary propargylic alcohols, Wang and coworkers subsequently reported the preparation of Schiff-base imino alcohols for the addition reaction of Zn-phenylacetylide with aromatic ketones. Ligand **17**, obtained by the condensation

of 1-naphthaldehyde with chiral amino alcohol that derived from L-phenylglycine, had been demonstrated to have improved catalytic activity (up to 1000 TON) and enantioselectivity in the addition reaction of Zn-phenylacetylide with aromatic ketones, offering the desired tertiary propargylic alcohols in up to 83% yield and 95% ee (Scheme 10).¹⁸



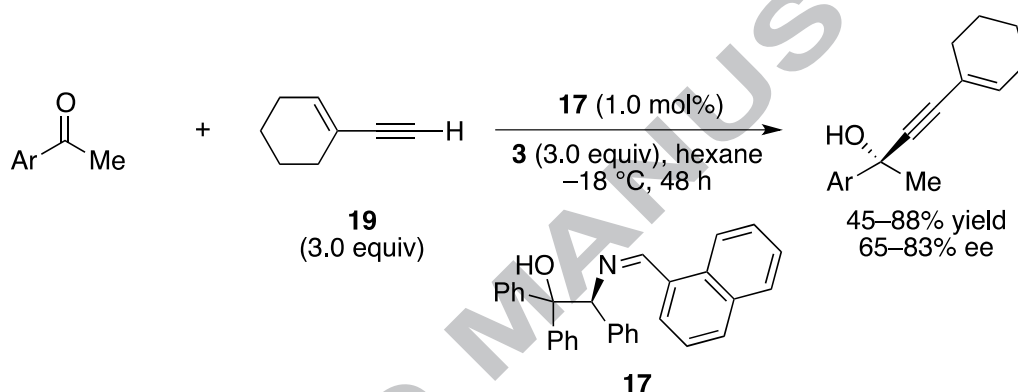
Scheme 10. Catalytic enantioselective addition of Zn-phenylacetylide to aromatic ketones in the presence of **17**.

Similarly, dipeptide **18** comprising L-valinol and *N*-Cbz protected L-phenylalanine was synthesized and reported as an efficacious ligand in the enantioselective addition of Zn-phenylacetylide to aromatic ketones, giving rise to tertiary propargylic alcohols in up to 90% yield and 91% ee (Scheme 11).¹⁹ In contrast to previous studies that hexane and toluene were optimal solvents, good chemical yields and enantioselectivities were obtained when the reactions were conducted in CH₂Cl₂.



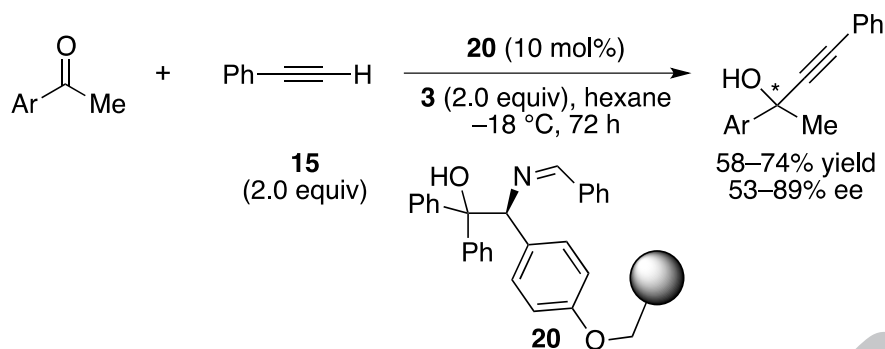
Scheme 11. Catalytic enantioselective addition of Zn-phenylacetylide to aromatic ketones in the presence of **18**.

The high catalytic activity of chiral Schiff base ligand **17** was later demonstrated in the catalytic addition of 1-ethynylcyclohexene (**19**) to aromatic and heteroaromatic ketones. The corresponding tertiary propargylic alcohols were isolated in 88% yield and 83% ee when the asymmetric reactions were carried out in the presence of 1 mol% of **17** (Scheme 12).²⁰ Interestingly, the reactions performed in CH₂Cl₂ yielded the racemic addition products. The asymmetric addition reaction saw good yields and enantioselectivities in the cases of heteroaryl ketones, such as 2-acetylfuran and 2-acetylthiophene.



Scheme 12. Catalytic enantioselective addition of Zn-phenylacetylide to aryl and heteroaryl ketones in the presence of **17**.

Having demonstrated the high catalytic reactivity and enantioselectivity of Schiff-base ligand **17** in the catalytic asymmetric alkynylation of ketones, Wang and coworkers sought to recover and reuse the chiral ligand for potential industrial application. The chiral ligand was immobilized on Merrifield resin, which facilitated the reuse of the catalyst by simple filtration. The catalytic addition of Zn-phenylacetylide with various aromatic ketones produced the corresponding tertiary propargylic alcohols in up to 74% yield and 89% ee in the presence of 10 mol% of polymer-supported iminoalcohol **20**. The polymer-supported chiral ligand remained catalytically active and selective in the successive experiments for five times (Scheme 13).²¹



Scheme 13. Catalytic enantioselective addition reaction of Zn-phenylacetylide to aromatic ketones in the presence of **20**.

The addition of dimethylzinc to α -ketoesters

The focus of this digest shifts to enantioselective addition reaction to α -ketoesters after introducing catalytic asymmetric addition of organozinc reagents to ketones. However, unlike ketones, α -ketoesters could coordinate with diorganozincs and enhance undesired background reaction that giving rise to a diminished stereoselectivity. Consequently, for this category of substrates, it is desirable to develop chiral catalysts that have higher catalytic activities to override the background reaction and/or to find reaction conditions that undesirable reaction pathway is completely suppressed.

Shibasaki and coworkers have developed a catalytic asymmetric addition of Me_2Zn (**13**) to methyl α -ketoesters. They anticipated that the nucleophilic Me_2Zn is strongly activated while an additional Lewis base is presence (Figure 2). Hence, *cis*-4-hydroxyprolinol **21** exhibited high catalytic activity and enantioselectivity, whereas the *trans*-counterpart gave the desired addition product without any selectivity. In the presence of 10 or 20 mol% of **21** and 27% of *i*PrOH as an additive, the addition reaction of Me_2Zn (**13**) with an array of methyl α -ketoesters proceeded in an enantioselective manner to generate the tertiary α -hydroxyl esters in up to 95% yield and 96% ee (Scheme 14).²² While detailed mechanistic study was impossible due to the absence of structural information of the active catalyst, the study of nonlinear effect²³ of this catalytic reaction revealed that positive nonlinear effect was

observed in the absence of *i*PrOH, whereas the nonlinearity was nearly vanished in the presence of *i*PrOH. This observation suggested that the active monomeric species was regenerated by adding a catalytic amount of *i*PrOH during the catalytic cycle, hence, improving the chemical yield and selectivity.

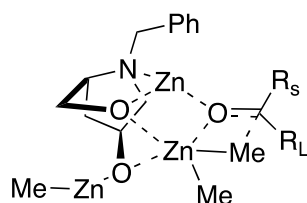
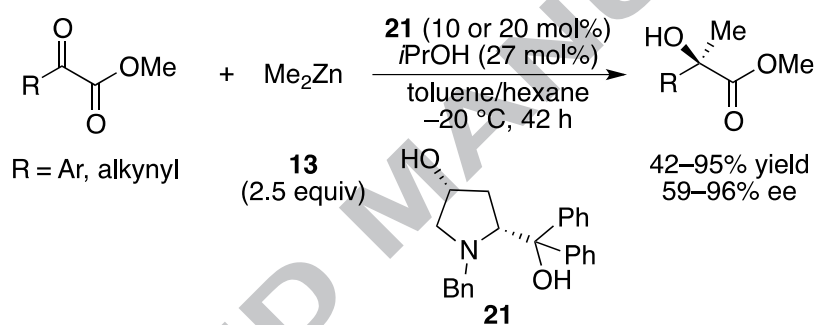
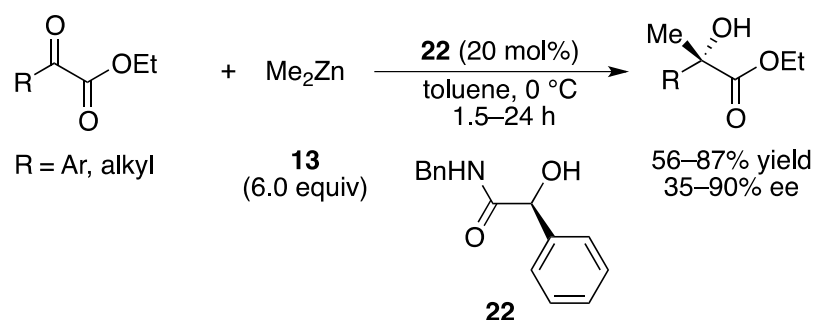


Figure 2. Operational model for *cis*-4-hydroxyprolinol-catalyzed addition reaction.



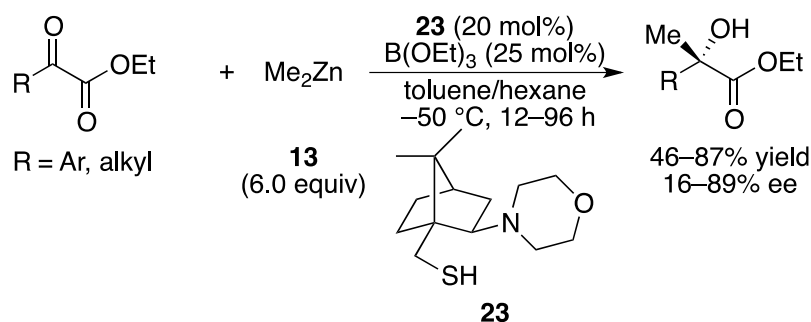
Scheme 14. Catalytic enantioselective Me_2Zn addition to α -ketoesters in the presence of **21**.

Pedro and coworkers reported that by tuning the electron-donating nature of the coordinating group of mandelamide ligand **22**, the asymmetric addition reaction of Me_2Zn (**13**) to ethyl α -ketoesters proceeded in a stereoselective manner with the tolerance of substrates containing heteroaryl groups (Scheme 15). The corresponding tertiary α -hydroxy esters were obtained in up to 87% yield and up to 90% ee without any additive.²⁴



Scheme 15. Catalytic enantioselective Me_2Zn addition to α -ketoesters in the presence of **22**.

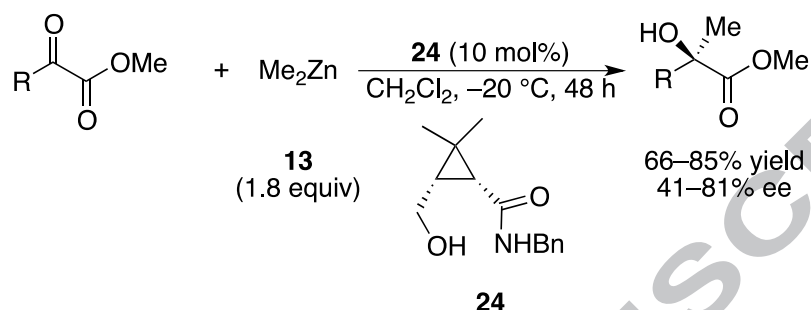
In the continuous efforts to develop efficient enantioselective transformations, we have synthesized a set of camphor-derived chiral aminothiols and realized that ligand **23** is an effective chiral modifier in the asymmetric syntheses of optically active alcohols via catalytic addition reactions of dialkylzincs, alkenylzincs and zinc acetylides with carbonyl compounds.²⁵ While the high catalytic activities and enantioselectivities of **23** were demonstrated in the enantioselective reactions with aldehydes, the addition of Me_2Zn (**13**) to ethyl α -ketoesters in the presence of 20 mol% of **23** was also highly selective to afford the corresponding chiral tertiary α -hydroxy esters in up to 87% yield and 89% ee when adding 25 mol% of $\text{B}(\text{OEt})_3$ as an additive (Scheme 16).^{25c} The reaction performed with and without catalytic amount of $\text{B}(\text{OEt})_3$ both showed positive nonlinear effect.



Scheme 16. Catalytic enantioselective Me_2Zn addition to α -ketoesters in the presence of **23**.

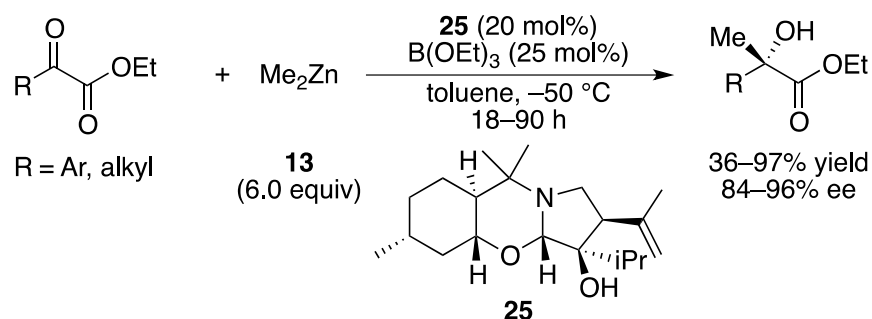
Zhong, Wang and coworkers synthesized a chiral amidoalcohol **24** as ligand with a

cis-substituted cyclopropane ring. In the presence of 10 mol% of **24**, asymmetric addition of Me₂Zn (**13**) to methyl α -ketoesters furnished tertiary α -hydroxyl esters in up to 85% yield and 81% ee without any additive (Scheme 17).²⁶



Scheme 17. Catalytic enantioselective Me₂Zn addition to α -ketoesters in the presence of **24**.

Chiral perhydro-1,3-benzoxazines, reported by Neito, Andrés and coworker, have shown high enantiocontrol in the Me₂Zn (**13**) addition reaction with ethyl α -ketoesters. In the presence of 20 mol% of **25**, the asymmetric addition proceeded to produce chiral tertiary α -hydroxy esters in 36–97% yield and 84–96% ee. In this reported method, 25 mol% of B(OEt)₃ was added to observe improved reactivity and selectivity (Scheme 18).²⁷ Presumably, reaction of B(OEt)₃ with Me₂Zn (**13**) generated B(CH₃)₃ and zinc ethoxide gave rise to enhancing the nucleophilicity of active nucleophilic species in the catalytic cycle, and therefore diminished the undesired background reaction.



Scheme 18. Catalytic enantioselective Me₂Zn addition to α -ketoesters in the presence of **25**.

In this digest, the recent development on the preparation of chiral alcohols with quaternary stereogenic center via catalytic asymmetric addition reaction of organozinc reagents to ketones is summarized. A variety of chiral ligands are synthesized and tested for the enantioselective alkylation, alkynylation and arylation to ketones and α -ketoesters. Based on these results, catalytic systems with higher catalytic activities and enantioselectivities will be developed and exploited in the syntheses of natural products and molecules with biological and pharmaceutical activities.

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Highlights:

1. Asymmetric synthesis of chiral *tert*-alcohols could be achieved with up to 99% ee.
2. Chiral ligand played an important role in the asymmetric catalysis.
3. Monomeric Zn-complex is proposed for the induction of high enantioselectivity.
4. Additive plays some role in the Zn-mediated asymmetric addition reaction.

Graphical abstract

